FLOOD DEPTH PRO

• • • •

1.1 | 8 2569

" — . "

0.1.4

/

: , , X.90, X.174, X.173A, X.44, Unit Hydrograph, Thiessen Polygon,

Flood Depth Pro 0.3.1 X.90 X.174
X.174
19–30 2568
581.20 . ./ 190.70 . .

- Q +1, +2 +3 Q ,
  API /
- Unit Hydrograph
- –
- X.90 X.174
  X.44

1–3	
4	
5	
6	Unit Hydrograph
7–8	Rating Table X.44/
9–11	
	Thiessen

1.

X.90 X.174

X.44

A. –	Q, API, X.90 7 X.174 3	Q X.90 X.174	
B.	X.90 + X.174 + + + —	(. ./)	
C. –	, DEM,	X.44,	

2.

1

	/	
Q	1 . . 2553–31 . . 2569; 30 . . 2567	, API, cross-correlation
–	baseflow	C, Unit Hydrograph,
Thiessen X.90	7 ; 1,534.59 . .	
X.90	1,534.31 . .	
Rating Table	X.90 X.173A 2568/69	H ↔ Q
DEM	DEM 2 .; 1,597	
X.44	25 . . 2568, H = 9.97 . .	
Thiessen X.174	3 ; 106.1931 . .	
Q WL X.174	2568/69 25 . . 2568	Rating Curve

1

2.1 Thiessen

Thiessen

$$\bar{P}(t) = \Sigma_i\, w_i\, P_i(t), \quad \Sigma_i\, w_i = 1 \quad (1)$$

$\bar{P}(t)$ t, $P_i(t)$ i w_i i

1,534.59 . .
1,534.31 . . 0.018%

2.2

Thiessen
0.50 IDW
R^2

X.173A

3.

3.1 (API)

API $\alpha = 0.88$

$$API(t) = \bar{P}(t) + 0.88 \times API(t-1) \quad (2)$$

API

3.2 lag

		lag	r
X.90	Q	2	0.536
X.90	ΔQ	1	0.444
X.173A	Q	2	0.582
X.173A	ΔQ	1	0.540

–

4.

$h = 1, 2, 3$ (direct-horizon)
80% 20%

$$\hat{Q}_s(t+h) = \max[0, \beta_0 + \beta Q\, Q_s(t) + \beta A\, \mathrm{API}(t) + \Sigma_{k=0}^{2} \beta Pk\, \bar{P}(t-k) + \Sigma_{j=1}^{h} \beta Fj\, \bar{P}f(t+j)] \quad (3)$$

$\hat{Q}_s(t+h)$ Q s . ./ ; $\bar{P}f$
max(0,·) Q

4.1 Ridge regression

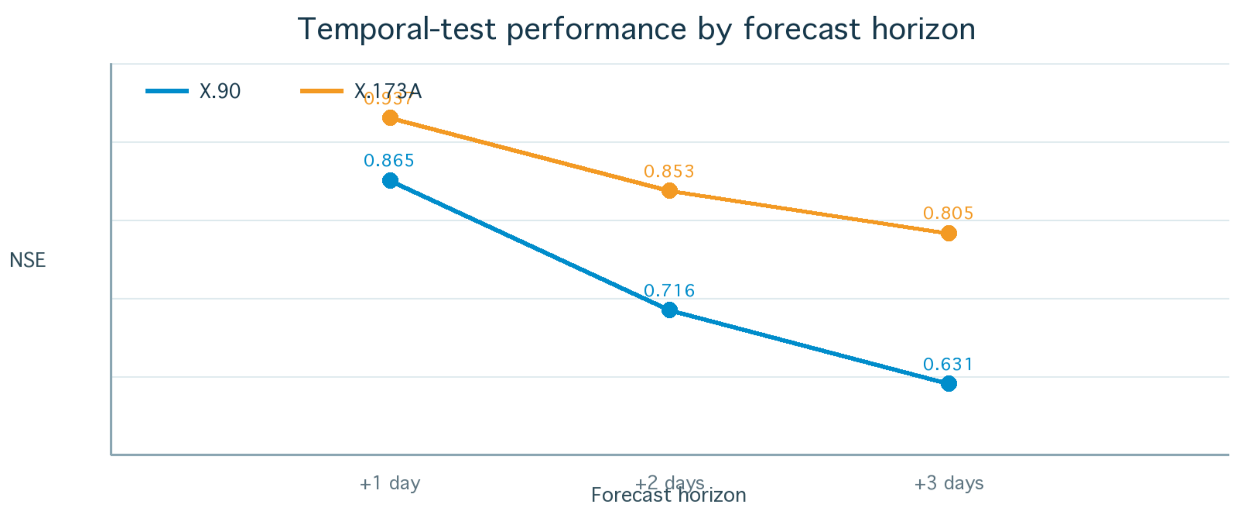
$\lambda > 0$

$$\hat{\beta} = \arg\min\beta \{ \Sigma_t[y_t - x_t^T\beta]^2 + \lambda \Sigma_j \beta_j^2 \} \quad (4)$$

λ X.90 10, 100 100 +1, +2 +3
X.173A $\lambda = 0$

β

4.2



1 NSE ;

		RMSE	MAE	R²	NSE	KGE	Bias
X.90	+1	17.20	10.73	0.869	0.865	0.873	−5.6%
X.90	+2	24.97	16.08	0.730	0.716	0.738	−11.4%
X.90	+3	28.45	18.70	0.653	0.631	0.669	−14.6%
X.173A	+1	6.51	3.85	0.938	0.937	0.930	−2.6%
X.173A	+2	9.94	6.15	0.858	0.853	0.852	−5.3%
X.173A	+3	11.46	7.27	0.812	0.805	0.811	−7.0%

2 (n = 991)

X.173A X.90 Bias

P̄f

5. Unit Hydrograph X.90

7

hydrograph

24

5.1

X.90 – direct runoff

45 ≥ 0.50 44

C

$$C_e = Rd,e / Pe \quad (5)$$

$$Pex(t) = C \times \bar{P}(t) \quad (6)$$

C	
25	0.082932
—	0.183038
75	0.280098
	44

C

5.2

baseflow lag

0–9 least squares

1

$$Rd(t) \approx \Sigma_{k=0}^{9} a_k \bar{P}(t-k), \quad a_k \geq 0; \quad u_k = a_k / \Sigma a_k \quad (7)$$

569 2010-11-20 2024-01-02 $R^2 = 0.587$ RMSE = 3.303 ./

5.3

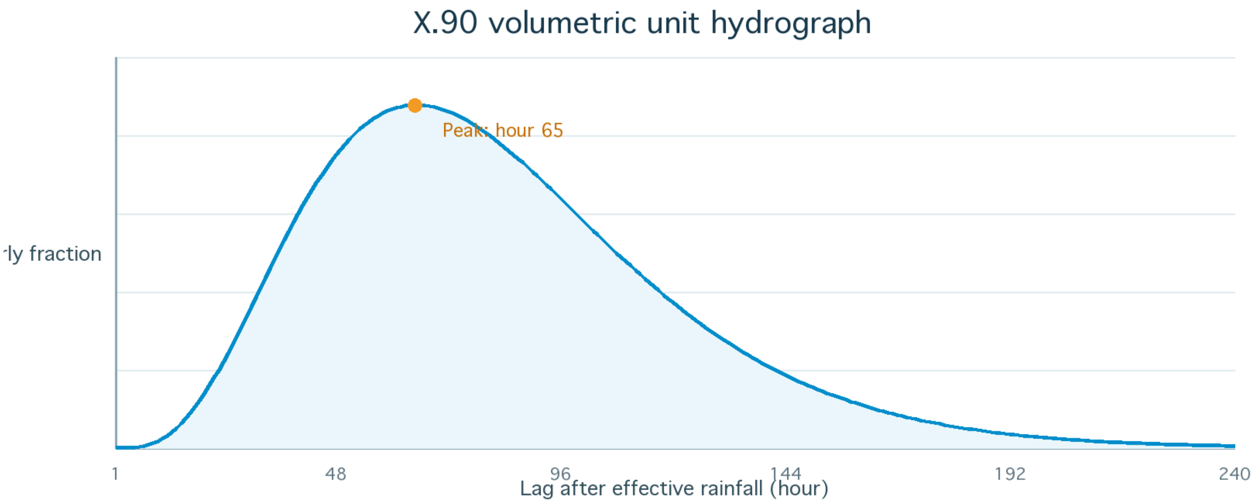
Gamma

240

1

$$g(\tau;k,\theta) = \tau^{k-1} \exp(-\tau/\theta) / [\Gamma(k)\, \theta^k] \quad (8)$$

k = 4.670664, θ = 0.734427 , 82.33 65



2 Unit Hydrograph X.90 (≈ 1)

5.4 convolution

$$V1mm = A \times 1{,}000 = 1{,}534{,}310\ m^3 \quad (9)$$

$$Vd(\tau) = \Sigma_t\ [Pex(t) \times A \times 1{,}000 \times u(\tau - 24t)] \quad (10)$$

$$Qtotal(\tau) = Qbase + Vd(\tau)/3{,}600 \quad (11)$$

$$Vday = \Sigma_{24}h\ Qtotal(\tau) \times 3{,}600\ /\ 10^6 \quad (12)$$

Q

Q

Gamma

hydrograph

telemetry

6. – Rating Table

H Q
2568/69

$$Q(H) = Q_a + [(H-H_a)/(H_b-H_a)](Q_b-Q_a) \quad (13)$$

	H (. .)	Q (. ./)	
X.90	2.00–13.05	0–5,110	1 . . 2568–31 . . 2569
X.173A	9.40–18.00	0–1,950	1 . . 2568–31 . . 2569
X.174	4.08–11.82	0.40–581.20	1 . . 2568–31 . . 2569

extrapolate Rating Table
H–Q backwater

7. X.44

X.90 X.174

10%

$$V_{main} = V_{X90} + V_{X174}; \quad V_{minor} = (r_{Rian} + r_{Wad} + r_{Tam})\, V_{main} \quad (14)$$

$$V_{in} = V_{main} + V_{minor} \quad (15)$$

$$V_{remain} = \max(0, V_{in} - V_{drain}) \quad (16)$$

7.1 – X.44

$$H_{X44} = 0.0097\, V_{remain} + 7.2383 \quad (17)$$

HX44 . . Vremain . ./
. . 2568

7.2383
Vremain ≤ 0.005 . ./

7.2

X.44	
7.40 . .	/
7.40–8.29 . .	
8.30–9.99 . .	
10.00 . .	

X.44 7.15 . .
7.40 . .

8.

dref DEM 2
residual 25 . . 2568 X.44 = 9.97 . .

$$f(H) = \max[0, (H-7.20)/(9.97-7.20)] \quad (18)$$

$$d(x,y,H) = dref(x,y) \times f(H) \quad (19)$$

–
DEM 2 .

	RMSE	⋆ 0.5 .
≤ 500 .	0.49 .	74%
≤ 1 .	0.52 .	72%
≤ 2 .	0.51 .	72%
≤ 4 .	0.52 .	70%

3 *9.97*
. .

(18)–(19)

9.

O_i , S_i , n

$$MAE = (1/n)\ \Sigma\ |S_i - O_i| \quad (20)$$

$$RMSE = \sqrt{[(1/n)\ \Sigma(S_i - O_i)^2]} \quad (21)$$

$$R^2 = [corr(S,O)]^2 \quad (22)$$

$$NSE = 1 - \Sigma(O_i - S_i)^2 / \Sigma(O_i - \bar{O})^2 \quad (23)$$

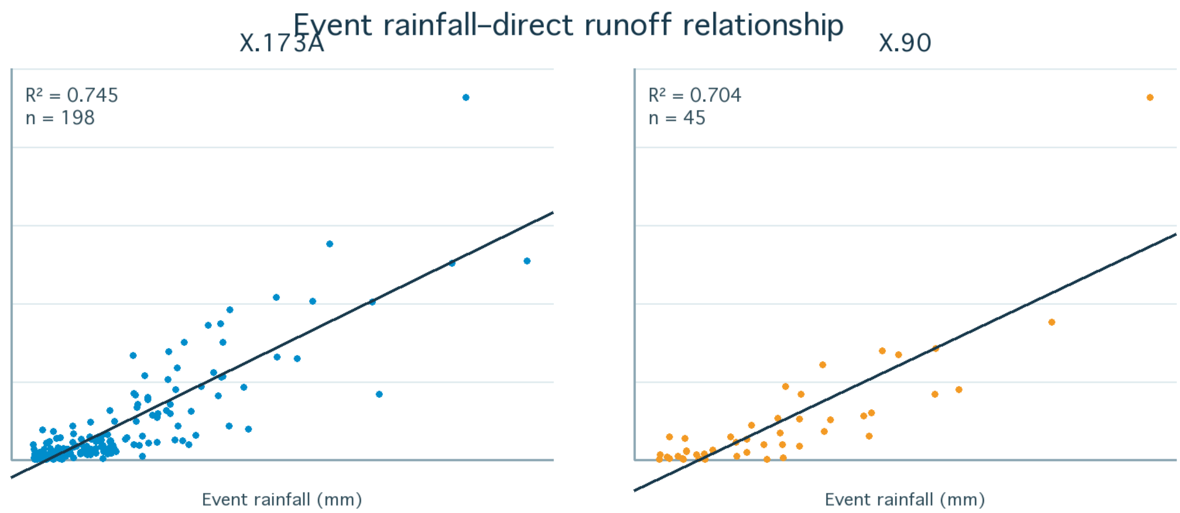
$$KGE = 1 - \sqrt{[(r-1)^2 + (\sigma S/\sigma O - 1)^2 + (\bar{S}/\bar{O} - 1)^2]} \quad (24)$$

$$Bias(\%) = 100 \times \Sigma(S_i - O_i) / \Sigma O_i \quad (25)$$

MAE/RMSE	0	Q; RMSE
R^2	1	bias
NSE	1 ; 0	
KGE	1	correlation, variability bias
Bias	0%	

10. –

Pe Rd baseflow X.90 – X.173A



3 –direct runoff depth

$$X.173A: Rd = 0.268385\ Pe - 8.791089;\ \ R^2 = 0.745;\ \ n = 198 \qquad (26)$$

$$X.90: Rd = 0.517669\ Pe - 27.185912;\ \ R^2 = 0.704;\ \ n = 45 \qquad (27)$$

	bootstrap 95% R^2		R^2 test	NSE train
X.173A	0.6518–0.8226	40	0.892	0.846
X.90	0.5647–0.8326	9	0.658	0.250

(26)–(27)

X.90 9

11.

	error
C baseflow	Unit Hydrograph C Qbase
Rating Table	/backwater
	42% 2568
	/
X.44	
	X.44 DEM
	trigger, uncertainty band real-time
X.174	580 411 580 232

11.1

1.
2. , Q , API, C, Qbase,
3. : – – C,
4. telemetry error /

5.

12.

1. X.90 7 X.174 3
2. Thiessen API;
3. / Q Rating Table
4. X.90 X.174

5. X.90 X.174

6. Vremain HX44 /
7.
8. error

13 X174

X.174 Thiessen 106.1931 . . 580 471, 580 411 580 232 QX.174 WLX.174 580 410 580 411 580 541 580 232

13.1

Thiessen 0.080464, 0.218715 0.700821 2568 580 232 580 411

580 232

$$\bar{P}_{174}(t) = 0.080464\ P_{580\,471}(t) + 0.218715\ P_{580\,411}(t) + 0.700821\ P_{580\,232}(t) \quad (28)$$

13.2 Rating Curve

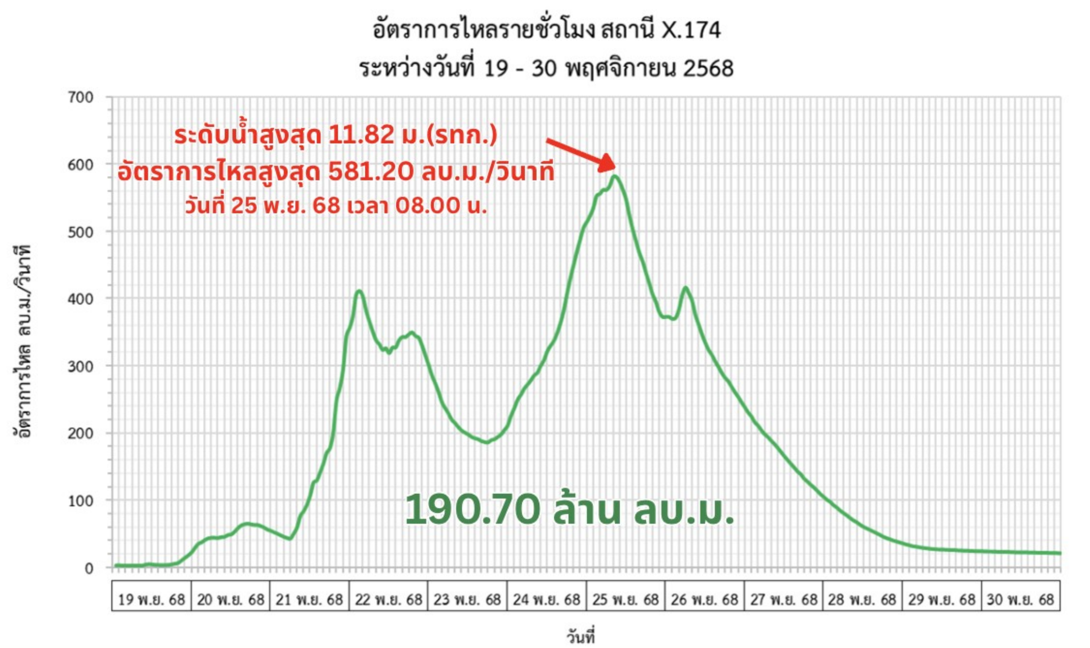
Rating Curve 2568 Q = a max(H − h0, 0)^b a = 10.37369 913, h0 = 3.91508 b = 1.800548 365 RMSE 5.564 . ./ , MAE 0.748 . ./ R^2 0.9831 4.08–11.82 . .

13.3 Unit Hydrograph

90 C 0.2621 25 0.1057 75 0.5063 Q 1 r = 0.5467 240

13.4 2568

19–30 2568 581.20 . ./ 08:00 . 25 190.70 . . Q 190.981 . . 0.15% 190.70 . .



4 Hydrograph X.174 19–30 2568

Gamma

Kv C

1.446767

$$Vd(\tau) = \Sigma t\ [Pex(t) \times A \times 1{,}000 \times Kv(t) \times u(\tau-24t)] \quad (29)$$

$$Kv(t) = 1 + [1.446767 - 1]\ fAPI(t), \quad 0 \le fAPI(t) \le 1 \quad (30)$$

fAPI 0 API 20 . 1 API 120 .
C 1.0 Kv
580 232

	581.20 . ./	580.20 . ./
	—	−0.17%
	190.70 . .	190.70 . .
NSE	—	0.822
RMSE	—	65.06 . ./

4 X.174 19–30 2568

13.5

X.174 validation
$R^2 = -0.6921$ persistence $R^2 = 0.4437$ Unit Hydrograph
replay

NSE, RMSE, bias

direct-horizon

X.90

h	λ	β₀	βQ	βAPI	βP0	βP1	βP2	βF1	βF2	βF3
+1	10	−0.100914	0.809316	−0.104257	1.223205	0.809335	0.049865	0.083430	—	—
+2	100	−0.759109	0.587910	−0.048327	1.636618	0.725130	−0.153650	1.191273	−0.012528	—
+3	100	−0.859828	0.471696	−0.054768	1.380619	0.347970	−0.251309	1.677020	1.129633	−0.023966

X.173A

h	λ	β₀	βQ	βAPI	βP0	βP1	βP2	βF1	βF2	βF3
+1	0	−0.507643	0.842621	−0.016023	0.540857	0.406920	−0.070213	0.057227	—	—
+2	0	−1.161378	0.648960	0.033261	0.801947	0.279399	−0.134265	0.588959	0.001049	—
+3	0	−1.478002	0.542196	0.041398	0.607901	0.141781	−0.160235	0.891766	0.554920	−0.022832

Thiessen X.90

				Lat	Lon	
Hat Yai (Khlong Hoi Khong)	HAT YAI AIRPORT	254.84	16.61%	6.87000	100.37000	name_match
Sa Dao (Tha Pho)	SA DAO	215.98	14.07%	6.79000	100.41000	coordinate_name_match
Agriculture Sa Dao	580254	448.30	29.21%	6.64000	100.42000	nearest_spatial_proxy
X.113	580221	224.73	14.64%	6.64000	100.39000	exact_code
Khlong Hoi Khong	580352	41.63	2.71%	6.90000	100.39000	name_match
X.90	580421	103.31	6.73%	6.93000	100.44000	exact_code
X.173A	580521	245.80	16.02%	6.82000	100.44000	not_available_idw_only

P_i, $\bar{P}$	i	./
API		.
Q, $\hat{Q}$		. ./
C		
Rd	baseflow	.
u(τ)	Unit Hydrograph τ	1/
V		. ./
H		. .
d		

.
2568 (33).

Hydro_X90_Analysis_Package. daily_basin_rainfall_runoff_X90.csv, event_runoff_coefficients.csv, rainfall_runoff_model_config.json analysis_report_X90_TH.md.

. Rating Table X.90 2025 (S.HC 2004/Y2025).

. Rating Table X.173A 2025 (S.HC 2010/Y2025).

Thiessen_X90_Hatyai_17-30Nov2025. Thiessen WGS84.

.

( ), 7 2569.

. X.174 2568/69.

Thiessen_X174. Thiessen WGS84
580 471, 580 411 580 232.

. Hydrograph X.174 19–30
2568.

Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I—A discussion of principles. Journal of Hydrology, 10(3), 282–290.

Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria. Journal of Hydrology, 377, 80–91.

1.1 Flood Depth Pro 0.3.1
X.174 2568
validation